

SEDAR 81 Gulf of Mexico Spanish Mackerel

Recommendations for considerations of future research are provided below and do not indicate any particular order of priority.

Composition data

- Further investigate seasonal migration patterns alongside the commercial fishing year (April-March) to ensure there are no errors in the data (see Stevens 2022a).
- Link the age data sets with the length only data sets to determine paired samples and better inform the appropriateness of utilizing CAAL.
- Implement systematic age sampling for the general recreational and commercial sectors. Sample sizes were limited, particularly for the shore sector, which account for the majority of the recent landings.

Develop defensible indices of relative abundance of Gulf of Mexico Spanish Mackerel

- Investigate whether the VL index could include “time fish” in the standardization process.
- Because this species is migratory, oceanic temperatures and circulation to which Spanish Mackerel respond vary spatially and temporally may impact availability to fishery participants. As such, it may be worth exploring an index which can account for spatiotemporal correlations.

Fishery Independent Age Data

- Age data from fishery independent (FI) sources are available but not incorporated in the model to date as they would need to be assigned to a new fleet whose selectivity would need to be specified. These are valuable data. Efforts should be made to figure out how to best incorporate FI age data into the assessment model.

Stock-recruit parameters

- Investigating the use of fixed vs. estimated parameters for steepness (h) and recruitment variability (sigmaR).

Shrimp Bycatch

- Estimates of shrimp bycatch are highly uncertain and need further investigation.

Gillnet fleet selectivity

- Investigating changes in catchability/selectivity for the Gillnet fleet.

Length composition of discards

- A better understanding of the size composition and mortality of discarded fish is needed, particularly for the recreational sector.

